

IMAGE COMPRESSION USING SINGULAR VALUE DECOMPOSITION (SVD)

**Under Supervision of
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Motivation & Problem Statement



Digital images consume large storage and bandwidth

Need for efficient, mathematically grounded compression

Key question:

How few singular values are needed for acceptable quality?

Application areas:

medical imaging, streaming, satellite imagery

What is ?

SVD

Singular Value Decomposition

U: left singular vectors ($m \times r$)

Σ : diagonal matrix of singular values (sorted descending)

V^T : right singular vectors ($r \times n$)

$$A = U \cdot \Sigma \cdot V^T$$

Singular values represent the "energy" of each component

Truncating to top k values gives the best rank-k approximation (Eckart-Young theorem)

Compression Pipeline

Step 1: Load image

convert to grayscale → represent as
2D matrix

Step 2

Apply full SVD decomposition

Step 3

Retain only top k singular
values/vectors

Step 4

Reconstruct approximated image

$$A_k = U_k \cdot \Sigma_k \cdot V_k^T$$

Step 5

Measure compression ratio and
PSNR

Step 6

Visualize results



Compression Ratio Formula

Original storage

$m \times n$ pixels

Compression Ratio

$(m \times n) / (k \times (m + n + 1))$

Higher k

Higher k $\rightarrow$ lower compression ratio $\rightarrow$ better quality

Compressed storage

$k \times (m + n + 1)$ values

Example

512×512 image at k=20 $\rightarrow$ CR $\approx$ 12.5×

Quality Metric: PSNR

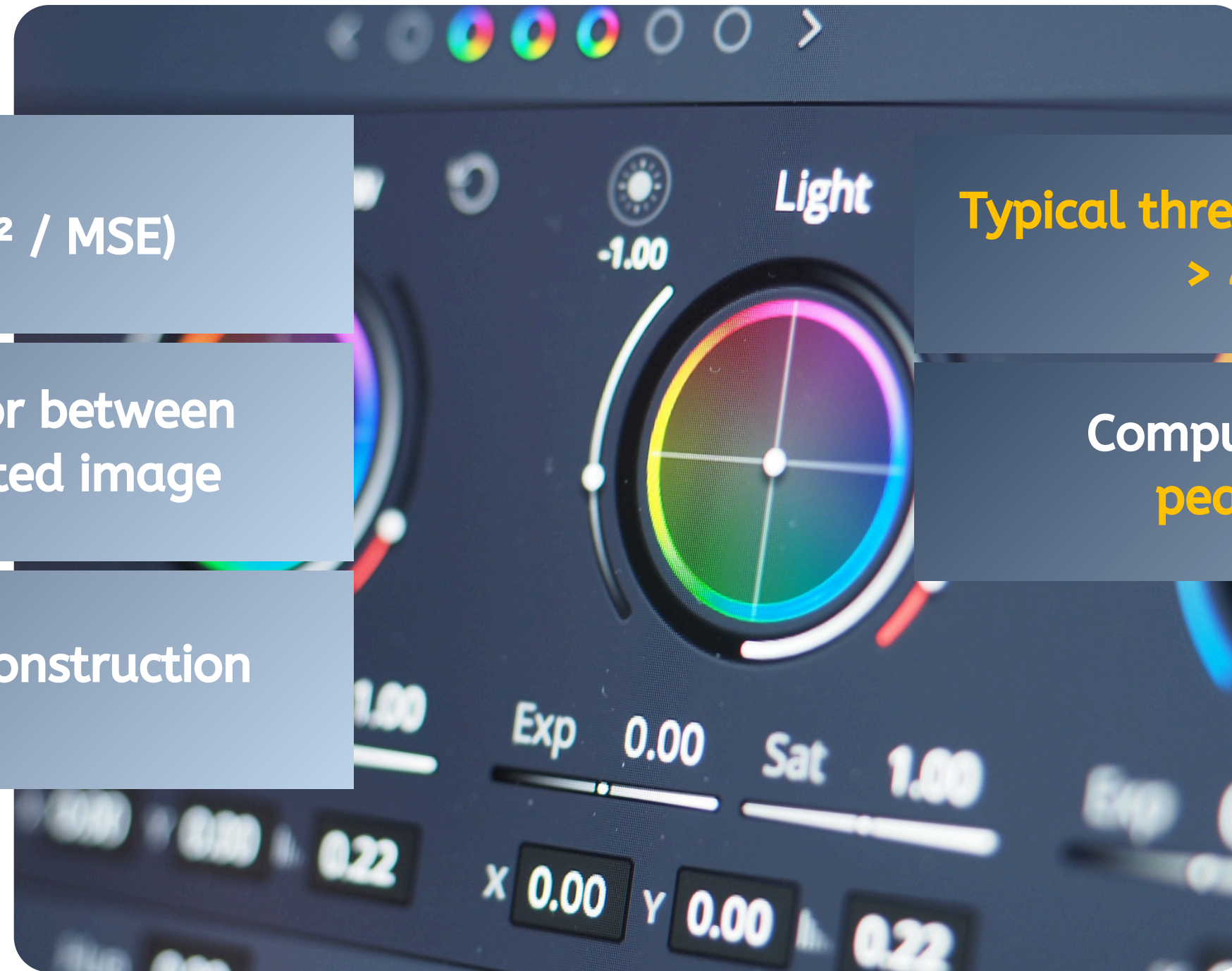
$$\text{PSNR} = 10 \cdot \log_{10}(255^2 / \text{MSE})$$

MSE: Mean Squared Error between original and reconstructed image

Higher PSNR = better reconstruction

Typical thresholds: > 30 dB = acceptable,
> 40 dB = high quality

Computed using `scikit-image's`
`peak_signal_noise_ratio`



Dataset

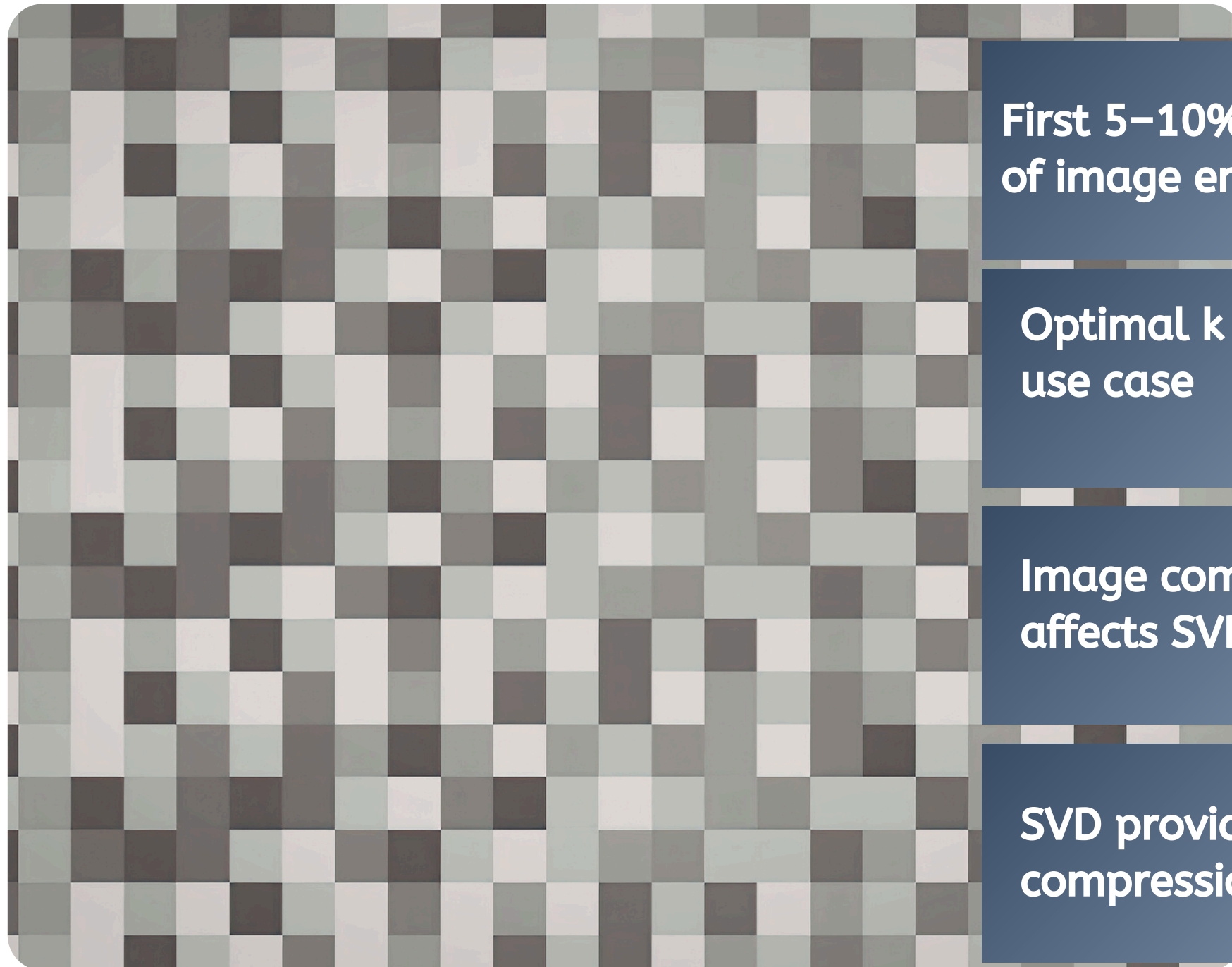
- Fixed built-in dataset : no upload required
- 5 scikit-image benchmarks: Camera, Coins, Moon, Clock, Page

All loaded automatically via skimage.dat

Converted to grayscale float64 arrays

k values tested: 5, 10, 20, 30, 50, 75, 100, 150, 200

Key Findings



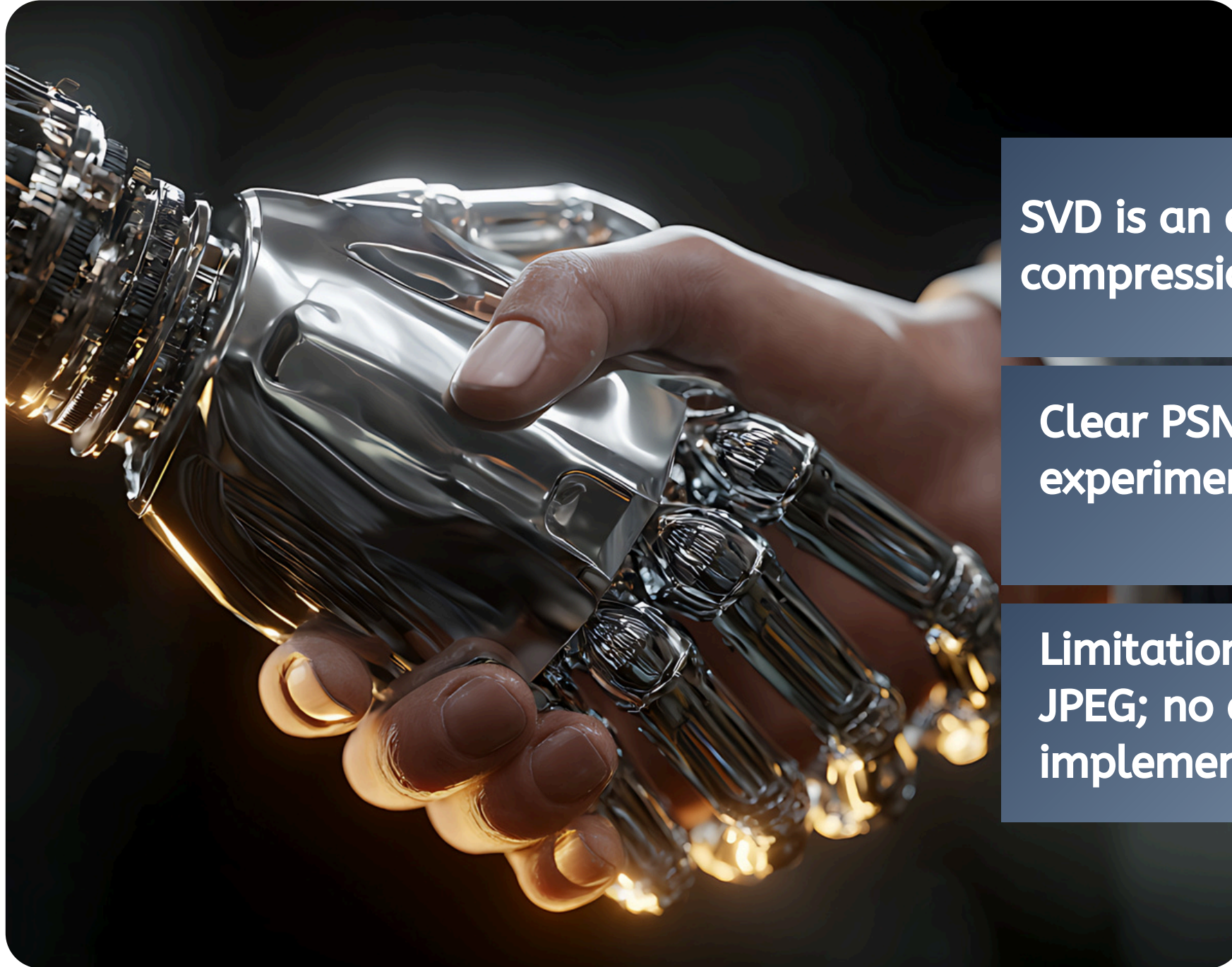
First 5–10% of singular values capture ~90% of image energy

Optimal k balances CR and PSNR based on use case

Image complexity (texture, edges) directly affects SVD efficiency

SVD provides a transparent, interpretable compression model

Conclusion



SVD is an effective and mathematically elegant compression method

Clear PSNR vs. CR trade-off quantified experimentally

Limitations: computationally heavier than JPEG; no color support in current implementation

Future Work



Extend to RGB channels

Compare with JPEG/DCT-based compression

Add SSIM as an additional quality metric



THANKYOU